

BECS 11431 ← Course code

Experiment No 00

2 cm
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INTRODUCTORY LABORATORY

Student Name :

Student Number :

Partner Name :

Partner Number

Date of Performance :

Date of Submission :

Date :

Experiment Number: 00

Experiment: Introductory Laboratory

APPARATUS:

1. Oscilloscope
2. Signal generator
3. Power supply
4. Analogue and Digital voltmeters
5. Resistors
 - $2 \times 100 \, \Omega$,
 - $220 \, \Omega$,
 - $1 \times 10^3 \, \Omega$
6. $0.047 \times 10^3 \, \text{F}$ capacitor
7. Connecting wires

Don't include figures

THEORY AND DIAGRAMS

- **Include the necessary theory which is used in the experiment**
- **Include the figures and circuit diagrams related to the experiment**
- **Draw the figures using a pencil**
- **Caption should be inserted**

PROCEDURE

- **Only include what you did in the lab**
- **Use Simple past passive voice.**

Examples -

- The power supply was switched on.
- The output voltage of the power supply was adjusted to 3V.
- The given circuit was created.
- Both the DVM and AVM were changed to DC measurements.
- The voltage V_2 was measured using both DVM and AVM.
- The theoretical value of V_2 was calculated using the voltage divider rule.
- The current through resistors was measured using DVM and AVM.
- The above values were compared with the theoretical current I .

OBSERVATIONS

Data

Experiment No

Student No

After taking the observation, you must get a signature from a demonstrator.

Do Not tipex



IN LAB CALCULATIONS

- If any calculations are needed while doing the experiment

CALCULATIONS

Calculate the experimental values under this part.

GRAPHS

1. Choosing the proper axis
2. Choosing a proper scale

The main divisions on the graph paper should be,

best 1,2,5,10,20,50,.....

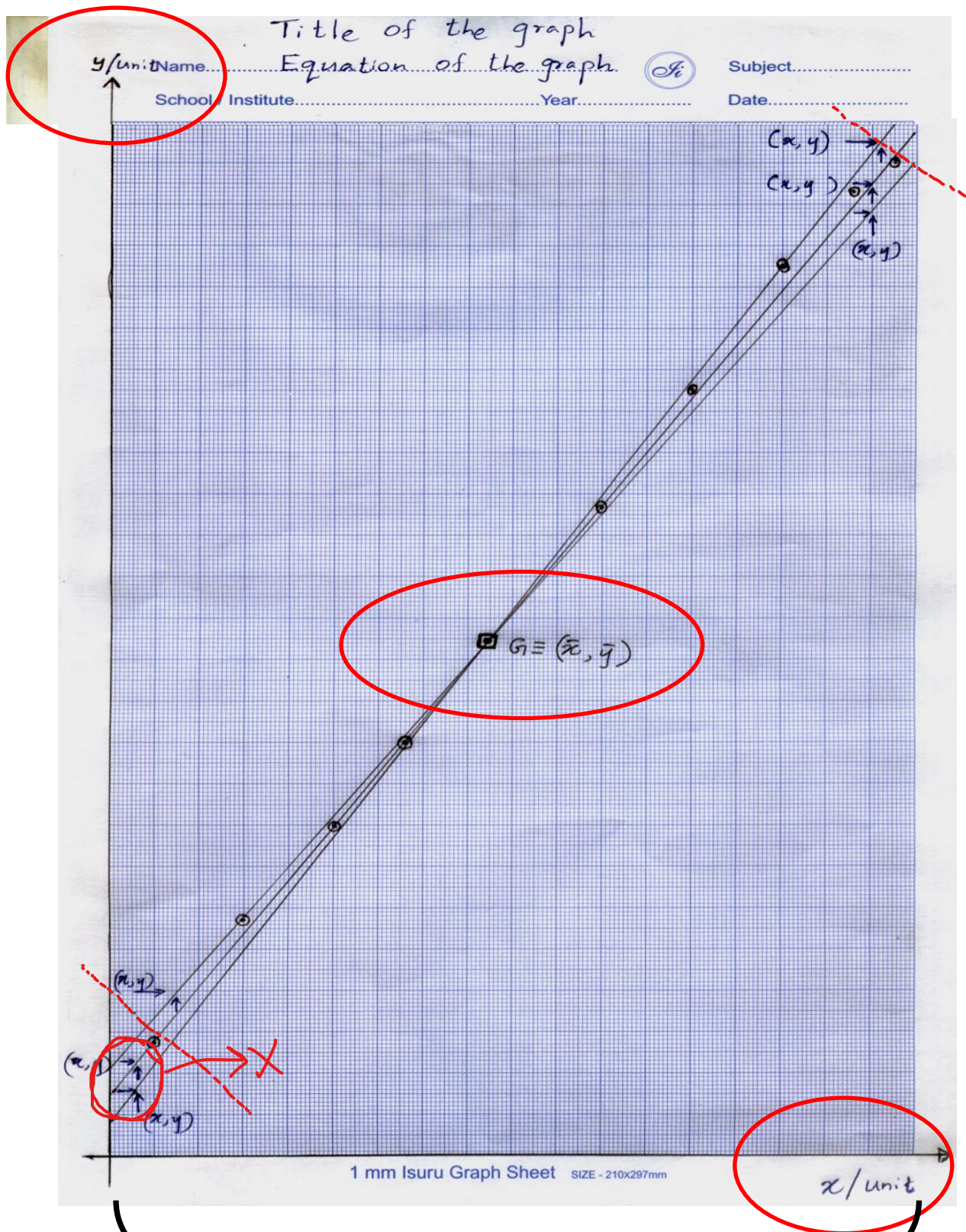
don't use 3,4,6,7,8,9,.....

3. Plotting graphs and calculations
4. Labeling the graph
5. Plotting error bars and calculations

- It is not necessary to choose the same scale for both axes.
- No need to start with Zero

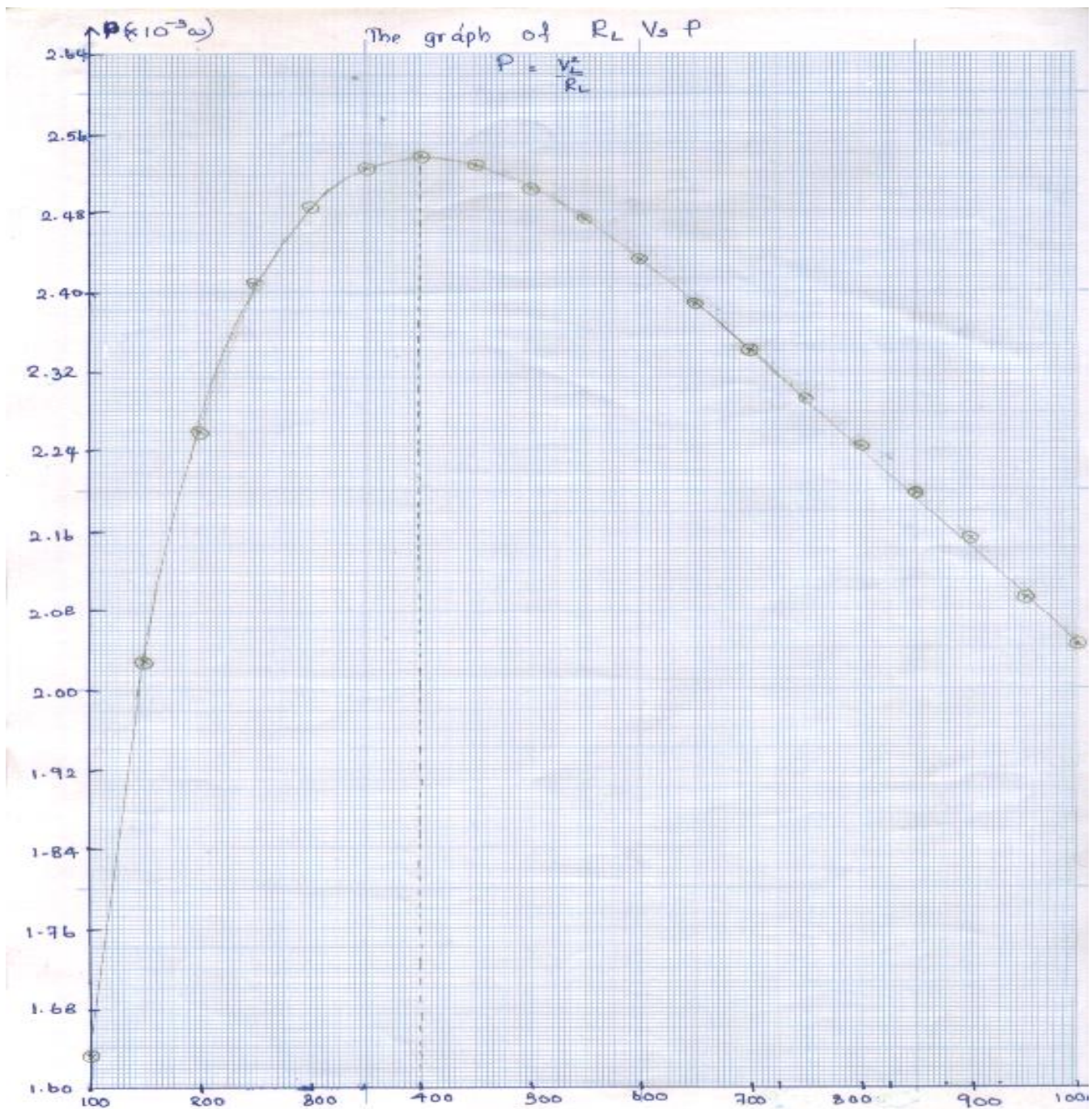
GRAPHS

The graph of Y vs X



Values should be marked On each axis

CURVE FITTING



Use a fixable Ruler to draw the curves

CONCLUSIONS

- **Make a table and compare the experiment values with the standard values**

Example -

Parameter	Standard Value (V)	Experimental Value (V)
Silicon Diode	0.7	0.68
Germanium Diode	0.3	0.28

DISCUSSIONS

Comparison with Theory

Discuss how the experimental findings align with the theoretical principles or predictions stated in the introduction. Highlight whether the experiment supports or refutes the theory and explains why

Sources of Error

Identify factors like equipment limits or environmental conditions that may have affected the results and explain their impact on measurements.

Suggestions for Improvement

If the experimental results align with the standard values, discuss the steps followed to achieve this accuracy.

If the results differ from the standard values, discuss the possible reasons for the deviation and suggest improvements to enhance accuracy.

Write the discussion in passive voice

QUESTIONS AND ANSWERS